linezolid-amr

Integrated AMR profiling + 23S rRNA
linezolid-resistance frequency analysis

User Guide · v0.2.1 · 2026-08-04

S. aureus · *E. faecalis* · *E. faecium* · *S. pneumoniae*

In-house MLST (PubMLST) → AMRFinderPlus → species-aware 23S read pileup. One command, one combined report, two summary CSVs (wide Kleborate-style + long per-feature), organism auto-detection, folder batch mode.

1 · Quick install

From bioconda recommended

```
conda create -n linezolid-amr -c bioconda -c conda-forge linezolid-amr
conda activate linezolid-amr
```

From source

```
git clone https://github.com/iowa69/linezolid-amr && cd linezolid-amr
conda create -n linezolid-amr -c bioconda -c conda-forge \
    python=3.11 ncbi-amrfinderplus minimap2 samtools bcftools mlst
conda activate linezolid-amr
pip install -e .
```

One-time setup (the tool auto-fetches if missing)

```
amrfinder -u                           # ~150 MB
linezolid-amr fetch-references         # ~120 KB total
```

2 · Pipeline overview

Step	Tool	Output
1. MLST & organism	in-house blastn vs bundled PubMLST schemes	Sequence Type + auto-detected organism
2. AMR profile	AMRFinderPlus	All AMR / virulence / stress genes & mutations
3. 23S heteroresistance	minimap2 + pysam + bcftools	Allele frequency at 18 curated LZD positions, per-allele evidence, full VCF

> **Why read-level for 23S?** *S. aureus* and *Enterococcus* spp. carry 4–6 copies of the 23S rRNA operon. A G2576T mutation present in only 1–3 copies produces a 15–60 % allele-fraction heteroresistant signal — invisible to assembly-only callers (consensus base stays G). Reading the raw reads is the only way to catch it.

> **Telling heteroresistance from artifact.** Because the signal is a *minority* allele it shares the frequency range with sequencing and alignment noise. Each candidate allele is therefore written to `<sample>.23S_lzd_evidence.tsv` with the evidence needed to separate the two: forward/reverse read counts and a Fisher exact strand-bias p-value (a real mutation is carried by the template and appears on both strands; an allele seen on one strand only is rejected), a Wilson 95 % confidence interval on the fraction, and the number of rrn operons that fraction implies. Each position also carries an **evidence tier** — substitutions only ever selected in archaea or mycobacteria are reported with their fractions but cannot drive a clinical positive unless you pass `--evidence-tier experimental`.

3 · Usage

3.1 · Single sample

```
linezolid-amr run \
    -a sample.fasta \
    -1 sample_R1.fq.gz -2 sample_R2.fq.gz \
    -o results/sample
```

MLST infers the organism. Override with `-O Enterococcus_faecium`. A warning prints if your override disagrees with MLST.

3.2 · Folder mode (batch)

```
linezolid-amr folder -i input_dir/ -o results/
```

Drops every `*.fasta/*.fa/*.fas/*.fna` into a sample, pairs it with FASTQs named with the same basename. Recognised read suffixes:

Pattern	Example
`_R1_001 / _R2_001`	S1_R1_001.fastq.gz / S1_R2_001.fastq.gz
`_R1 / _R2`	S1_R1.fq.gz / S1_R2.fq.gz
`_1 / _2`	S1_1.fastq.gz / S1_2.fastq.gz

Produces two cohort-level CSVs at `results/`: `ALL_samples.summary_wide.csv` (one row per sample) and `ALL_samples.summary_long.csv` (one row per gene/mutation).

3.3 · All options

Flag	Default	Meaning
`-a / --assembly`	—	FASTA assembly (run only)
`-1 / --r1`	—	Forward FASTQ (run only)
`-2 / --r2`	—	Reverse FASTQ (optional)
`-i / --input`	—	Input folder (folder only)
`-o / --outdir`	—	Output directory
`-s / --sample`	assembly stem	Sample name
`-O / --organism`	auto (MLST)	Override MLST organism inference
`-t / --threads`	all CPUs	Parallel threads
`--plus`	off	AMRFinderPlus stress/virulence/biocide search
`--min-af`	0.15	Min 23S alt-allele frequency for a positive call
`--min-depth`	20	Min depth at a 23S position
`--min-baseq`	13	Discard bases below this Phred quality

`--min-mapq`	20	Discard reads below this mapping quality
`--min-alt-reads`	3	Min reads supporting an allele before it can be called
`--strand-bias-p`	1e-3	Fisher exact p below which strand distribution counts as biased
`--no-strand-filter`	off	Report strand-biased alleles as positive calls (pre-0.2 behaviour)
`--evidence-tier`	associated	Weakest evidence tier allowed to drive a POSITIVE call
`--reads-only`	off	(folder) process every R1/R2 pair without assemblies
`--skip-amrfinder / --skip-rrna23s`	—	Skip one step

4 · Output structure

```
results/sample/
├── amrfinder/amrfinder.tsv              # all AMR / virulence / stress hits
├── rrna23s/
│   ├── sample.23S.bam (+ .bai)          # sorted, indexed alignment
│   ├── sample.23S.vcf.gz (+ .csi)       # all 23S variants
│   └── sample.23S_lzd_pileup.tsv        # AF at canonical LZD sites
├── sample.linezolid_amr.json            # combined machine-readable report
├── sample.linezolid_amr.txt             # plain-text summary
├── sample.summary_wide.csv              # 1 row · AMR class buckets as cols
└── sample.summary_long.csv              # 1 row per gene/mutation
```

In **folder** mode each sample sits in its own subfolder, plus the two cohort CSVs at the root of `outdir/`.

summary_wide.csv columns (fixed prefix)

Column	Meaning
`sample`	sample name
`organism`	AMRFinderPlus organism used
`mlst_scheme / ST`	MLST scheme + sequence type
`linezolid_call`	`POS` if any LZD signal, else `neg`
`lzd_23S_mutations`	e.g. `G2576T:0.6647(173)` — pos:af(depth)
`lzd_genes`	e.g. `cfr(D);optrA;23S_G2576T`
`AMR_<class>`	one column per AMRFinderPlus class — `gene(cov/ident)`
`VIRULENCE / STRESS_<class>`	only present with `--plus`

5 · Worked example — Enterococcus faecium VRE LZD-R

```
$ linezolid-amr run -a test_assembly.fasta -1 test_R1.fq.gz -2 test_R2.fq.gz \
                    -o results/test_efaecium -s test -t 8

=== test ===
>> MLST / organism inference...
   organism: Enterococcus_faecium   MLST scheme: efaecium   ST: 17
>> Running AMRFinderPlus...
   17 hits, 3 linezolid-relevant
>> Running 23S rRNA analysis...
   18 positions; 1 with resistance allele

Linezolid resistance call: POSITIVE
```

Linezolid signal

Source	Detection
AMRFinderPlus	`23S_G2576T`, `cfr(D)`, `optrA`
23S pileup at E. coli 2576	G=58, T=115 (AF=**0.665**) — ~4/6 operoni mutati

summary_wide.csv excerpt

```
sample,organism,mlst_scheme,ST,linezolid_call,lzd_23S_mutations,lzd_genes,AMR_OXAZOLIDI
NONE,AMR_GLYCOPEPTIDE,AMR_BETA-LACTAM,...
test,Enterococcus_faecium,efaecium,17,POS,G2576T:0.6647(173),"cfr(D);optrA;23S_G2576T",
"cfr(D)(100/100.0);optrA(100/99.7);23S_G2576T(100/100.0)","vanA(100/99.7);vanH-
A(100/100.0);vanX-A(72/100.0);vanY-A(100/99.7);vanZ-
A(100/100.0)","pbp5_E629V(100/95.4);pbp5_M485A(100/95.4)",...
```

Heteroresistance read-out. E. faecium has 6 rrn operons. 0.665 ≈ 4/6 mutated copies — a typical clinical heteroresistance pattern. The tool quantifies it exactly; conventional callers would just report *G2576T present* without the operon fraction.

6 · References

All canonical 23S positions in `loci.json` carry a verified PMID. Full bibliography in `CITATION.cff`.

PMID	Reference	Role
10556031	Kloss et al. 1999, JMB	Foundational 23S mutational mapping
11476839	Tsiodras et al. 2001, Lancet	First clinical G2576T description
22143525	Long & Vester 2012, AAC	Comprehensive resistance review
16801432	Long et al. 2006, AAC	Cfr methyltransferase mechanism
25977397	Wang et al. 2015, JAC	optrA discovery
29635422	Antonelli et al. 2018, JAC	poxtA discovery